C R RAO Advanced Institute of Mathematics, Statistics and Computer Science (AIMSCS)
in Collaboration with Jawaharlal Nehru Technological University Hyderabad
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
IV B. Tech I Sem Project Stage–I – Review–II Evaluation & Request Form

MAJOR PROJECT STAGE 1 – REVIEW 2

Student Guidelines (From Project Idea to a Technically Defined Solution)

1. Purpose of Review 2: The **Second Review of Major Project Stage 1** is an important milestone in your project development. Review 1 focused mainly on understanding the problem, introducing the project, studying the initial literature, identifying the research gap, and proposing a possible solution.

At this **Review 2** stage, you are expected to demonstrate that you have moved beyond the project idea and have started converting your idea into a **clear, technically feasible, and implementable solution**.

The expected progression is:

Review 1: Problem → Literature → Research Gap → Proposed Methodology (Optional)
Review 2: Refined Problem → Detailed Methodology → Dataset/Data → Implementation Plan → Initial Progress

Therefore, Review 2 should demonstrate **real progress**, not simply a revised version of the Review 1 presentation.

2. What Should Be Achieved by Review 2? : By the end of Review 2, every project team should have clarity about:

- What exactly is the final problem?
  - The problem statement should be refined based on literature study and Review 1 feedback.
- What exactly are you proposing?
  - The proposed methodology should be sufficiently detailed to explain how the problem will be solved.
- What data will you use?
  - The dataset, data source, input characteristics, and preprocessing requirements should be identified.
- How will you implement the solution?
  - The tools, technologies, algorithms, models, modules, and implementation steps should be clearly identified.
- How will you evaluate your solution?
  - You should know what metrics, baseline methods, and experiments will be used.
- What progress have you made?
  - There should be evidence of technical progress wherever applicable.

3. Review 2 Is NOT Review 1 Again: One of the most common mistakes is preparing Review 2 as a slightly modified version of Review 1.

Avoid spending most of your presentation explaining:

- What is Artificial Intelligence?
- What is Machine Learning?
- What is Blockchain?

- What is Deep Learning?
- General background of the domain.
- The same literature review presented in Review 1.

A small amount of background may be retained for continuity, but the **focus should now be your project**.

Your presentation should answer:

> **"What have you done after Review 1, and how are you going to implement the proposed solution?"**

4. Expected Structure of Review 2: Your **Review 2** presentation should preferably follow the sequence below.

4.1 Section 1 – Project Overview

Briefly present: Project title, Team members, Supervisor, Problem domain, One-slide project overview. Do not spend excessive time on basic introduction.

4.2 Section 2 – Review 1 Feedback and Changes

This section is very important.

Clearly mention:

Feedback Received

What suggestions or corrections were given during Review 1?

Actions Taken

What did you do based on those suggestions?

For example:

Review 1 Suggestion	Action Taken
Study more recent papers	Added recent research papers
Improve problem statement	Problem statement refined
Compare existing methods	Comparison table prepared
Identify suitable dataset	Dataset finalized
Clarify methodology	Methodology redesigned

This demonstrates that you are **actively responding to faculty guidance**.

4.3 Section 3 – Refined Problem Statement

Present the final/refined problem statement. Your problem statement should clearly communicate:

Existing Situation → Problem → Limitation → Need → Proposed Direction

It should be specific and technically meaningful.

Avoid broad statements such as:

"The objective of this project is to develop an AI system."

Instead, explain exactly:

- What problem exists?
- Why are existing solutions insufficient?
- What specific challenge are you addressing?
- What type of solution are you proposing?

4.4 Section 4 – Refined Objectives

Present **3–5 clear objectives**.

Objectives should directly support your problem statement.

For example:

1. To collect and preprocess the required dataset.
2. To develop a baseline model for the identified problem.
3. To propose an improved methodology for addressing the identified research gap.
4. To evaluate the proposed approach using appropriate performance metrics.
5. To compare the proposed approach with existing baseline methods.

Your objectives should be **specific and achievable**.

4.5 Section 5 – Updated Literature Review

The literature review should now be more focused than Review 1.

Include:

- Important papers studied after Review 1.
- Recent research.
- Relevant methodologies.
- Comparative analysis.
- Limitations of existing approaches.
- Refined research gap.

A useful comparison table can contain:

Paper	Year	Method	Dataset	Performance	Limitation
Paper 1	2026	Method A	Dataset X	92%	Limitation A
Paper 2	2025	Method B	Dataset X	94%	Limitation B
Paper 3	2024	Method C	Dataset Y	93%	Limitation C
Proposed	2026	Method D	Dataset X/Y	To be evaluated	Addresses identified gaps

The purpose is not to show many papers. All team members should download, read and present the papers as per the requirement.

The purpose is to demonstrate:

"We understand the existing solutions and know where our proposed work fits."

4.6 Section 6 – Final Research Gap: The research gap should now be much clearer than it was during Review 1.

Avoid statements such as:

"There is a lack of research in this area."

Instead explain:

- What existing approaches do well.
- Where they fail.
- Why that limitation matters.
- What has not been adequately addressed.
- How your project will attempt to address it.

The research gap should logically lead to your proposed methodology.

Important Relationship

Your presentation should show:

Literature Review → Limitations → Research Gap → Proposed Methodology

If these four elements do not logically connect, your project requires further refinement.

4.7. Section 7 – Detailed Proposed Methodology: This is one of the **most important sections of Review 2**. Your methodology should no longer be only a high-level block diagram.
Explain each major component.
For example:
Input Data
↓
Data Collection
↓
Preprocessing
↓
Feature Extraction / Representation
↓
Baseline Model
↓
Proposed Model
↓
Prediction
↓
Evaluation
↓
Comparison and Analysis
Depending on your project, additional modules may be included.
Explain Every Major Component
For each component, explain:
What?
What does the component do?
Why?
Why is it required?
How?
How will it be implemented?
For example: **Data Augmentation**
What? Generates additional training samples.
Why? Helps address limited training data and improve model generalization.
How? Apply suitable transformations such as rotation, scaling, cropping, or flipping.
This level of explanation demonstrates genuine understanding.
4.8. Clearly Identify Your Proposed Contribution
One of the most important questions during Review 2 will be:
"What is new in your project?"
You should be able to answer this clearly. Your contribution may involve:

- A new architecture.
- A hybrid methodology.
- A modified algorithm.
- A new feature representation.
- A new preprocessing strategy.

- An optimization technique.
- An ensemble approach.
- An explainability component.
- A new application of an existing technique to a specific problem.

Do not claim novelty merely because you combined two technologies.
You should explain **why the combination is meaningful and what problem it addresses**.
4.9 Section 8 – Dataset / Data Collection: At Review 2, the data source should preferably be identified.
Present:

- Dataset name.
- Dataset source.
- Number of samples.
- Number of classes/categories.
- Input features or image dimensions, where applicable.
- Data distribution.
- Training/validation/test split.
- Data preprocessing requirements.

If the dataset is publicly available, mention its source.
If you are collecting your own data, explain:

- Data collection method.
- Sample size.
- Collection procedure.
- Ethical/privacy considerations were applicable.
- Data labeling process.

4.10 Section 9 – Baseline Method: Where applicable, implement or at least define a **baseline approach**.
A baseline provides a reference point for evaluating your proposed methodology.
For example:

Existing CNN → Baseline
Proposed CNN + Attention → Proposed Approach

Then you can compare:

Model	Accuracy	Precision	Recall	F1-Score
Baseline	—	—	—	—
Proposed	—	—	—	—

The baseline helps demonstrate whether your proposed method provides an improvement.

4.11 Section 10 – Initial Implementation Progress: This is a major difference between Review 1 and Review 2. Students should demonstrate **evidence of progress**. Depending on the project, evidence may include:
Software/AI/ML Projects

- Dataset downloaded.
- Dataset preprocessing completed.
- Exploratory data analysis.
- Initial code developed.
- Baseline model implemented.
- Training started.

- Initial results obtained.
- Sample predictions.
- Confusion matrix.
- Graphs/plots.

Application Development Projects

- System architecture.
- Database design.
- UI prototype.
- Login/module implementation.
- Backend development.
- API implementation.
- Initial integration.

Hardware/IoT Projects

- Component identification.
- Circuit/block diagram.
- Sensor integration.
- Initial prototype.
- Data collection.
- Communication testing.

Cybersecurity Projects

- Dataset preparation.
- Environment setup.
- Feature extraction.
- Baseline implementation.
- Initial detection results.

Do not show fabricated results.

If implementation has not yet produced reliable results, clearly state:

"Implementation is in progress."

Honest preliminary progress is better than unsupported results.

4.12 Section 11 – Tools and Technologies: Clearly identify the technologies required.

For example:

Software

- Python
- Java
- MATLAB
- R
- SQL
- etc.

Frameworks/Libraries

- TensorFlow
- PyTorch
- Scikit-learn
- OpenCV
- Pandas
- NumPy
- etc.

Development Environment

- Google Colab
- Jupyter Notebook
- VS Code
- Android Studio
- Eclipse
- etc.

Do not list technologies simply because they are popular.

Every technology listed should have a **specific role in your project**.

4.13 Section 12 – Evaluation Strategy : Your evaluation plan should be clear.

Explain:

What will you measure?

For example:

- Accuracy.
- Precision.
- Recall.
- F1-score.
- MAE.
- RMSE.

- mAP.
- Response time.
- Memory usage.
- Processing time.

What will you compare?

For example:

Proposed Method vs Existing Method

or

Proposed Model vs Baseline Model

What will demonstrate success? You should define the expected improvement or performance criterion.

4.14 Section 13 – Current Status: Present your project status honestly.

A simple status table can be used:

Activity	Status
Problem identification	Completed
Literature review	Completed
Research gap	Completed
Methodology design	Completed
Dataset identification	Completed
Data preprocessing	In Progress
Baseline implementation	In Progress
Proposed model	Yet to Start
Evaluation	Yet to Start

This helps the review committee understand your actual progress.

4.15 Section 14 – Work Completed After Review 1

This section should answer one direct question:

"What did you do after Review 1?"

Include:

- Papers studied.
- Problem refinement.
- Methodology changes.
- Dataset selection.
- Software/environment setup.
- Coding completed.
- Experiments performed.
- Faculty feedback incorporated.

The review committee should be able to clearly see **progress between Review 1 and Review 2**.

5. Questions Every Student Must Be Prepared to Answer: Before Review 2, every student should be able to answer:

Problem

1. What exactly is your problem?
2. Why is it important?
3. What is the scope of your project?

Literature

4. What are the most important papers you studied?
5. What methods are commonly used?
6. What are their limitations?
7. What research gap did you identify?

Methodology

8. What exactly are you proposing?
9. Why did you select this methodology?
10. What is the role of each component?
11. What is your expected contribution?

Data

12. Which dataset are you using?
13. Why did you select this dataset?
14. How will you preprocess the data?

Implementation

15. What have you implemented so far?
16. What tools and technologies are you using?
17. What difficulties have you encountered?

Evaluation

18. How will you evaluate your system?
19. What baseline will you use?
20. How will you demonstrate improvement?

If every team member can answer these questions confidently, the team is likely to be well prepared.

6. Common Mistakes in Review 2 :

Mistake 1 – Repeating Review 1

Do not present the same content again with minor changes.

Review 2 must demonstrate progress.

Mistake 2 – No Technical Detail

Saying:

> "We will use deep learning."

is not enough.

Explain:

> Which model? Why? What input? What preprocessing? What architecture? What improvement?

Mistake 3 – No Dataset

Students should not reach Review 2 without knowing where their data will come from, unless the project genuinely requires data collection that is still underway.

Mistake 4 – No Implementation Progress

A project cannot remain at the idea stage indefinitely. Start implementation early.

Mistake 5 – Unsupported Novelty

Do not say:

> "This is a new method."

unless you have compared it with existing research and can explain the contribution.

Mistake 6 – One Student Knows Everything

Every team member should understand the project.

During the review, faculty may ask questions to any team member.

Mistake 7 – Showing Results Without Understanding Them
Never display a graph or accuracy value that you cannot explain.
You should know:

- How the result was generated.
- What it represents.
- Why it changed.
- Whether it is reliable.

7. Recommended Review 2 PPT Structure (May include additional content as per internal guide suggestions)
A practical PPT structure is:

1. **Title Slide**
2. **Project Overview**
3. **Review 1 Feedback and Actions Taken**
4. **Refined Problem Statement**
5. **Objectives**
6. **Updated Literature Review**
7. **Comparative Analysis**
8. **Research Gap**
9. **Proposed Contribution**
10. **Detailed Proposed Methodology**
11. **System/Architecture Diagram**
12. **Dataset/Data Collection**
13. **Tools and Technologies**
14. **Baseline Approach**
15. **Initial Implementation Progress**
16. **Preliminary Results, if available**
17. **Evaluation Strategy**
18. **Current Status**
19. **Remaining Work and Timeline**
20. **Expected Final Outcome**
21. **References**

The exact number of slides may be adjusted according to the Supervisor instructions.
8. A Simple Test Before Attending Review 2
Ask yourself these five questions:
1. Problem
Can I explain my problem in one minute?
2. Gap
Can I explain what is missing in existing research?
3. Method
Can I explain exactly how my proposed solution will work?
4. Progress
Can I show what we have actually done after Review 1?
5. Contribution
Can I explain what we expect to contribute through this project?

If the answer to all five questions is **YES**, you are on the right track.

Final Message:

- Review 2 is the stage where your project should begin moving from **an idea into an actual technical project**. Do not wait for the final stage to start implementation. Do not treat the review as a formality.
- Use the review as an opportunity to receive feedback **before you invest significant time in the wrong direction**.
- Your internal guide will help you improve your methodology, and make your project stronger.

Therefore:

- **Come prepared.**
- **Know your problem.**
- **Understand your literature.**
- **Know your research gap.**
- **Explain your methodology.**
- **Show your progress.**
- **Be ready to defend your decisions.**
- **Dress neatly and maintain a professional appearance during the project review.**

Remember: Most importantly, remember:

- **Review 1 gave you the direction.**
- **Review 2 should demonstrate that you are moving in that direction.**
- **The final project should demonstrate that you reached the intended destination.**

Your Review 2 Goal

By the end of Review 2, you should confidently be able to say:

> **"We have refined the problem, strengthened our literature study, finalized our proposed methodology, identified the required data and tools, started implementation, and have a clear plan to complete and evaluate our solution."**

Take Review 2 seriously. The quality of your implementation and final project depends on the decisions you make at this stage.

If the Project Review Committee is not satisfied with your presentation, preparation, or progress, an additional review may be scheduled. Ensure that your work, implementation, documentation, and presentation are completed to the expected standard.